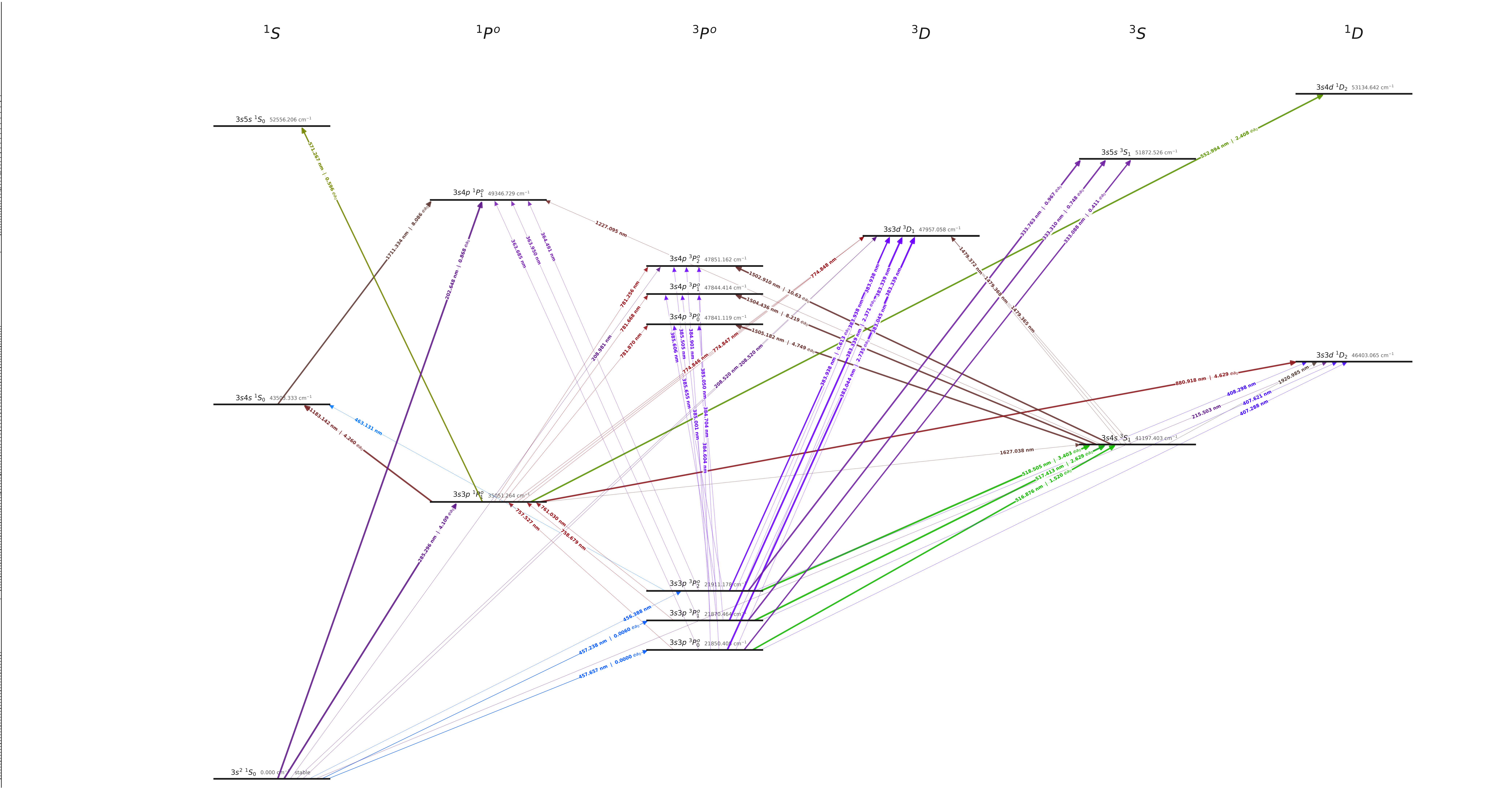


Magnesium-24

^{24}Mg (Mg I), nuclear spin $I = 0$. Energy levels and transitions below 2 μm : 18 levels, 61 lines.

Mg $^{+}$ ionisation limit 61671.05 cm^{-1} = 7.64624 eV (162.151 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e r J' \rangle $ (ea $_0$)	A (s $^{-1}$)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3s^2\ ^1S_0 - 3s4p\ ^1P_1^o$	202.6477	202.5825	1479.3777	0.868	6.120e+07	-	-	-
$3s^2\ ^1S_0 - 3s3p\ ^1P_1^o$	285.2964	285.2126	1050.8105	4.109	4.910e+08	-	-	-
$3s3p\ ^3P_1^o - 3s5s\ ^3S_1$	333.3104	333.2146	899.4392	0.748	1.020e+07	-	-	-
$3s3p\ ^3P_2^o - 3s5s\ ^3S_1$	333.7634	333.6674	898.2186	0.967	1.700e+07	-	-	-
$3s3p\ ^3P_0^o - 3s3d\ ^3D_1$	383.0441	382.9355	782.6578	2.735	8.990e+07	-	-	-
$3s3p\ ^3P_1^o - 3s3d\ ^3D_1$	383.3387	383.2299	782.0564	2.371	6.740e+07	-	-	-
$3s3p\ ^3P_0^o - 3s4s\ ^3S_1$	516.8761	516.7321	580.0084	1.520	1.130e+07	-	-	-
$3s3p\ ^3P_1^o - 3s4s\ ^3S_1$	517.4125	517.2684	579.4071	2.629	3.370e+07	-	-	-
$3s3p\ ^3P_2^o - 3s4s\ ^3S_1$	518.5048	518.3604	578.1865	3.403	5.610e+07	-	-	-
$3s3p\ ^1P_1^o - 3s4d\ ^1D_2$	552.9940	552.8405	542.1260	2.408	1.390e+07	-	-	-
$3s3p\ ^1P_1^o - 3s3d\ ^1D_2$	880.9175	880.6756	340.3184	4.629	1.270e+07	-	-	-
$3s3p\ ^1P_1^o - 3s4s\ ^1S_0$	1183.1423	1182.8185	253.3867	4.260	2.220e+07	-	-	-
$3s4s\ ^3S_1 - 3s4p\ ^3P_2^o$	1502.9099	1502.4992	199.4747	10.63	1.350e+07	-	-	-
$3s4s\ ^3S_1 - 3s4p\ ^3P_0^o$	1504.4356	1504.0246	199.2724	8.219	1.340e+07	-	-	-
$3s4s\ ^3S_1 - 3s4p\ ^3P_0^o$	1505.1817	1504.7705	199.1736	4.749	1.340e+07	-	-	-
$3s4s\ ^1S_0 - 3s4p\ ^1P_1^o$	1711.3336	1710.8663	175.1806	8.086	8.810e+06	-	-	-

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea $_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

